

JORGE MARINEZ DIAZ

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EDUCATION

California State University, East Bay

Bachelor of Science, Computer Science | December 2024

Bachelor of Science, Biochemistry | December 2024

Relevant Coursework: Biochemistry I & II, Organic Chemistry I & II, Major Organ Chemistry, Physical Chemistry I & II, Calculus III, Principles of Physics II

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C++, PHP

Web Technologies: React, HTML, CSS

Scientific Computing: NumPy, Data Analysis, Machine Learning

Tools: Git, GitHub

Laboratory Techniques: HPLC, PCR, ELISA, SDS-PAGE, Gel Electrophoresis

EXPERIENCE

Coding Instructor | Code With Us | July 2019–Present

- Teach Python, Java, C++, JavaScript, HTML, and CSS to students of varying experience levels.
- Guide students through debugging, algorithms, software design, and project development.
- Develop customized lesson plans and technical exercises to strengthen programming fundamentals.

PROJECTS

Artificial Neural Network | Python, NumPy

GitHub: github.com/JorgeMarinez/Artificial-Neural-Network

- Built a feedforward neural network from scratch in Python using NumPy, implementing forward propagation, backpropagation, and gradient-based optimization without TensorFlow or PyTorch.
- Trained the network for XOR classification, achieving correct predictions across all input combinations after 10,000 training epochs.

PPO Activity Analysis via HPLC

- Developed and validated an HPLC calibration model ($R^2 = 0.9997$) to quantify chlorogenic acid concentrations in apple peel extracts.
- Analyzed chromatographic data and measured chlorogenic acid concentrations ranging from 0.89–115.57 mg/100 g to evaluate the effects of thermal treatment on PPO activity.

Process Scheduler Simulator | Python

GitHub: github.com/JorgeMarinez/processScheduler

- Implemented Round Robin, Shortest Job First (SJF), and Shortest Remaining Time First (SRTF) CPU scheduling algorithms in Python.
- Simulated process execution, preemption, turnaround time, and waiting time statistics to model operating system scheduling behavior.

Library Management System | Java

GitHub: github.com/JorgeMarinez/LibraryManagementSystem

- Developed a Java-based library management application using object-oriented design principles and multi-class architecture.
- Implemented functionality for managing books, patrons, inventory records, and lending transactions.